Data Science and Artificial Intelligence

Machine Learning

Linear Discriminant Analysis

Lecture No. 1

By- SIDDHARTH SABHARWAL SIR

Recap of Previous Lecture

- Topic
- Topic
- Topic
- Topic
- Topic

SVM → Regression
→ Hinge loss fxn

Topics to be Covered

- Topic
- Topic
- Topic
- Topic
- Topic

linear discriminant analysis

YOU ARE
STRONGER
THAN YOU
THINK

Hinge Loss function

$$\max(0, 1 - y_i(wx_i + b))$$

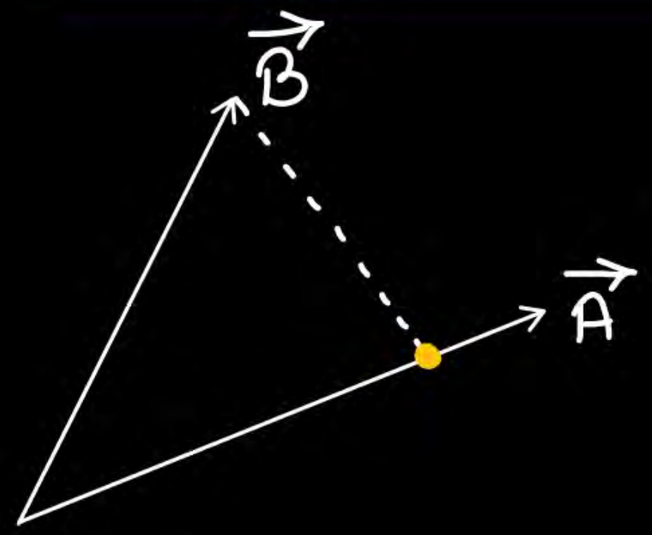
REVISION

SVM can be used for regression also

$$\min \frac{1}{2}\|w\|^2$$
$$\text{st} \quad |y-\hat{y}| \leq \varepsilon_0$$

$$\min \frac{1}{2}\|w\|^2 + C\sum \alpha_i$$
$$\text{st} \quad |y-\hat{y}| \leq \varepsilon_0 + \alpha_i$$

Projection of one vector on other



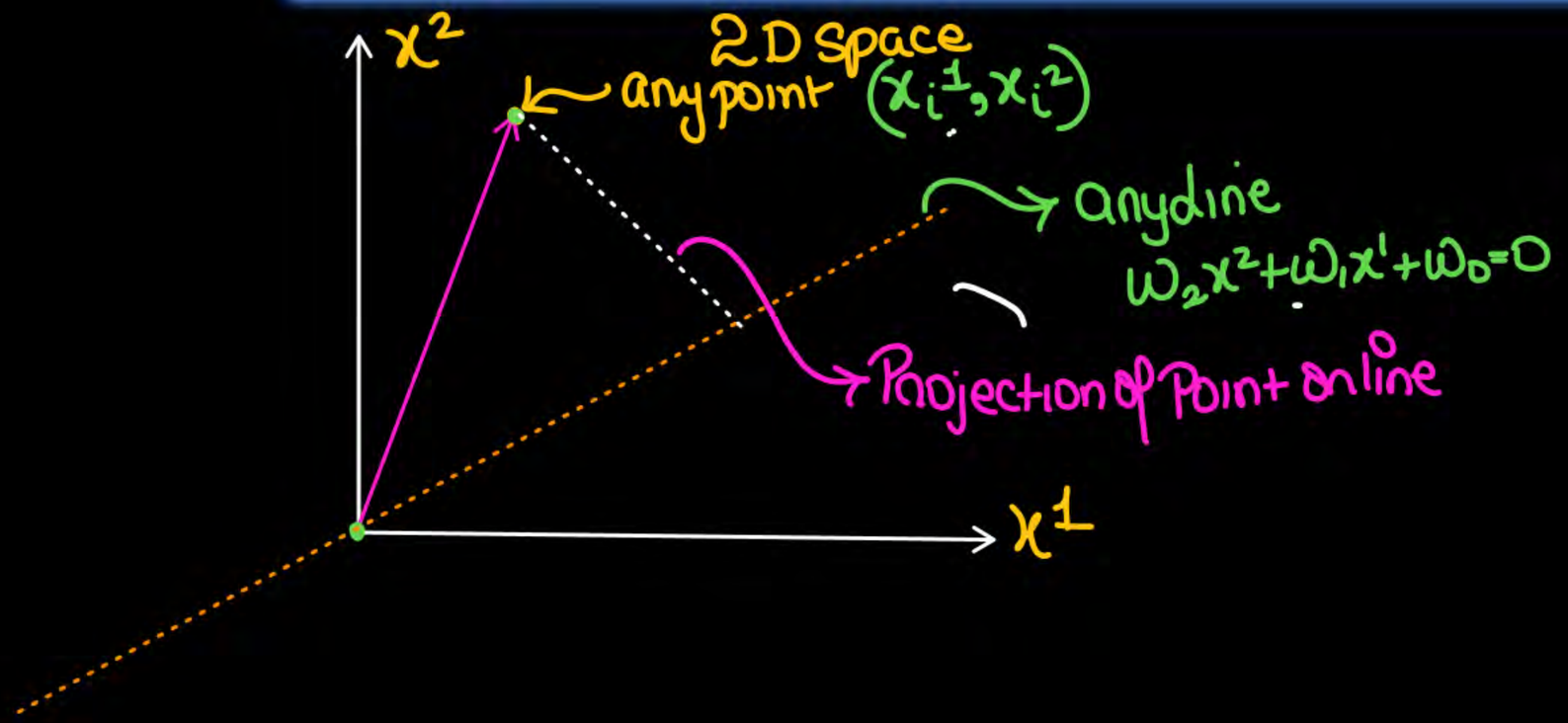
- Projection of vector B on A
  - ↳ dot product

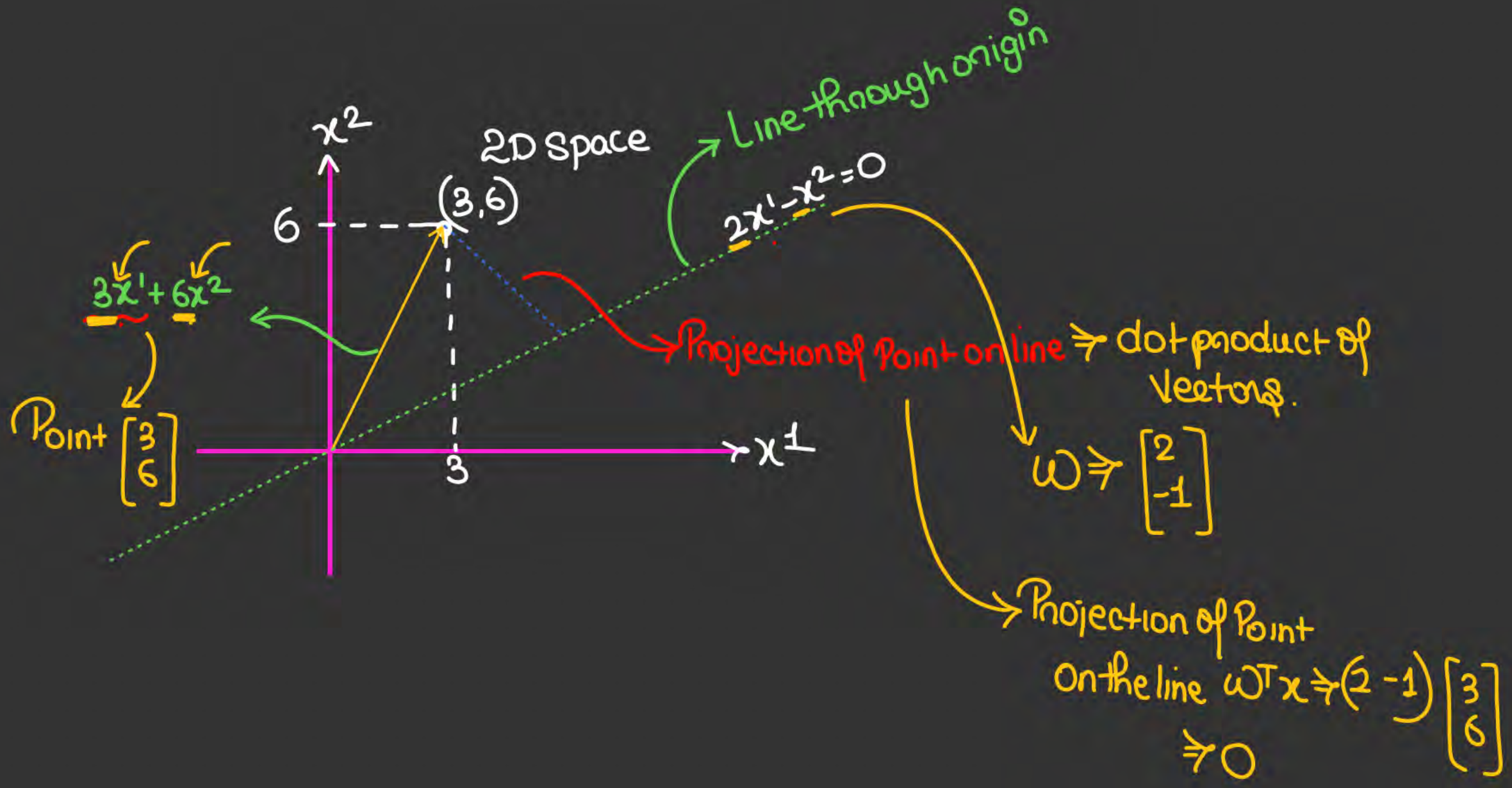
$$\vec{A} = A_x \hat{a_x} + A_y \hat{a_y} + A_z \hat{a_z}$$

$$\vec{B} = B_x \hat{a_x} + B_y \hat{a_y} + B_z \hat{a_z}$$

$$\Rightarrow \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Projection of one vector on other




2D space
x^2
x^1
6
3
(3,6)
$3x^1 + 6x^2$
Point $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$
Line through origin
$2x^1 - x^2 = 0$
Projection of Point on line ⇒ dot product of vectors.
$w \Rightarrow \begin{bmatrix} 2 \\ -1 \end{bmatrix}$
Projection of Point on the line $w^T x \Rightarrow (2\ -1) \begin{bmatrix} 3 \\ 6 \end{bmatrix}$
$\Rightarrow 0$

So if we have any point $x_i^0 = \begin{bmatrix} x_i^1 \\ x_i^2 \\ \vdots \\ x_i^D \end{bmatrix}$

D dimension

and any plane $w_1x^1 + w_2x^2 + \dots w_Dx^D = 0$ → Passing through origin

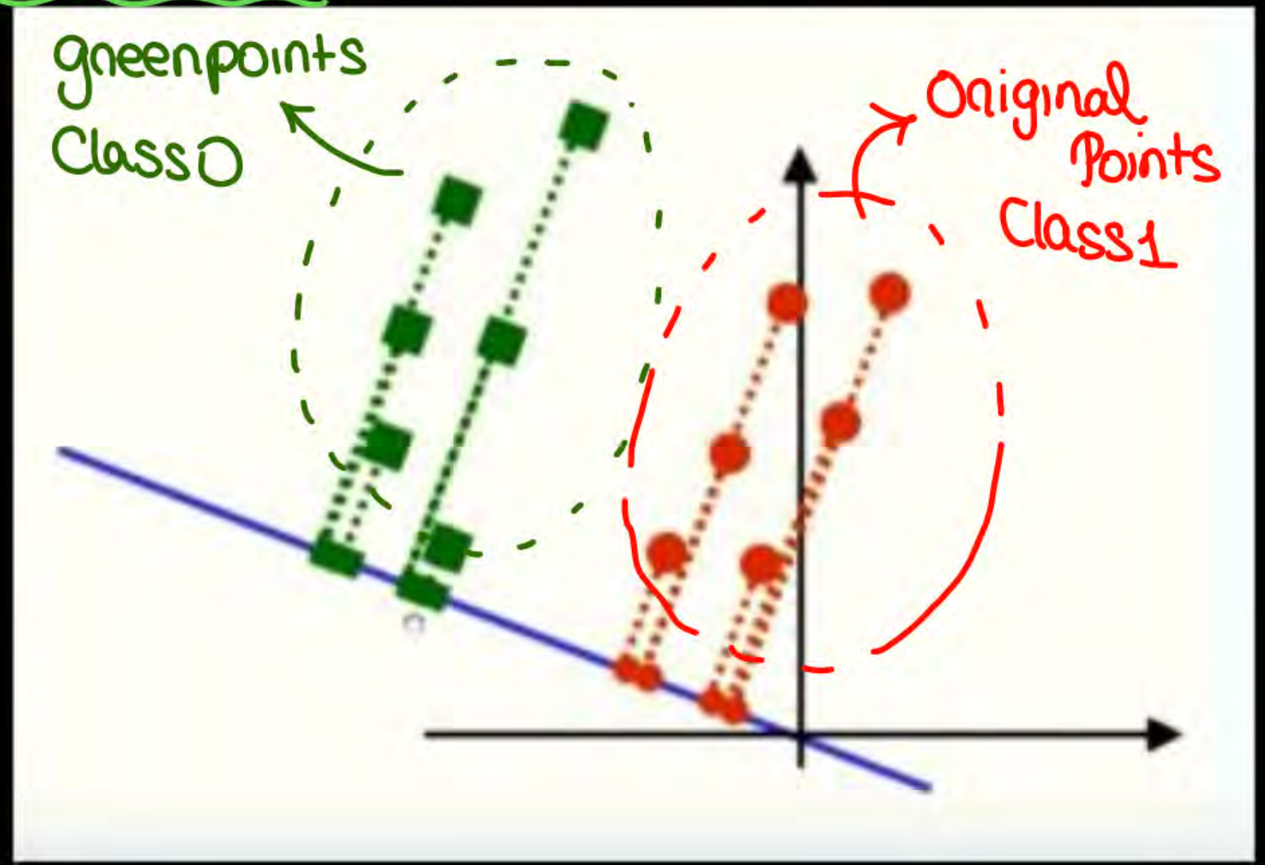
$\Rightarrow W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{bmatrix}$

So Projection of any Point $\Rightarrow w^T x_i^0$

What is a LDA ?

- The main task in LDA is to project the data onto a line or a plane such that the distance between the points is maximized.

⇒ LDA → dimensionality Red and Classification



- So in LDA we project the points of different classes onto a plane,

⇒ Can be used for multiple Classes

→ Such that the projected points Can be linearly Classified.

So we need that the projected points of different classes shd be Seperated from each other

What is a discriminant function

- LDA does this by projecting data with two or more dimensions into one dimension so that it can be more easily classified. The technique is, therefore, sometimes referred to as dimensionality reduction.
- Linear Discriminant Analysis (LDA) is a statistical method used for classification and dimensionality reduction. It finds the linear combination of features that best separates two or more classes of objects or events.

What is a discriminant function

- This versatility ensures that LDA can be used for multi-class data classification problems.

In LDA ⇒ the projection shd be such that

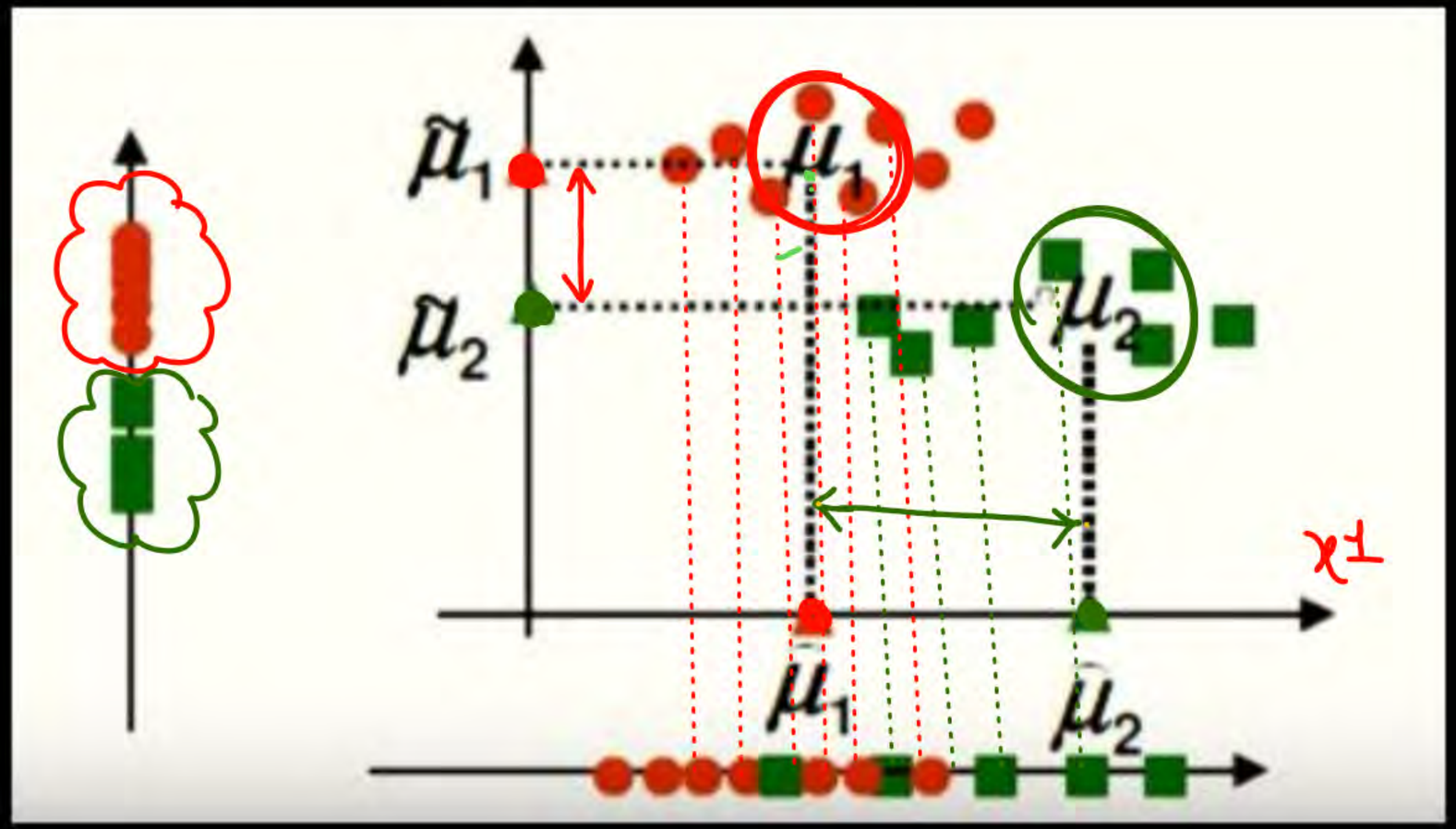
1) the points of different classes are Seperated from each other

→ thus we want that projected mean location of different classes shd be far away from each other.

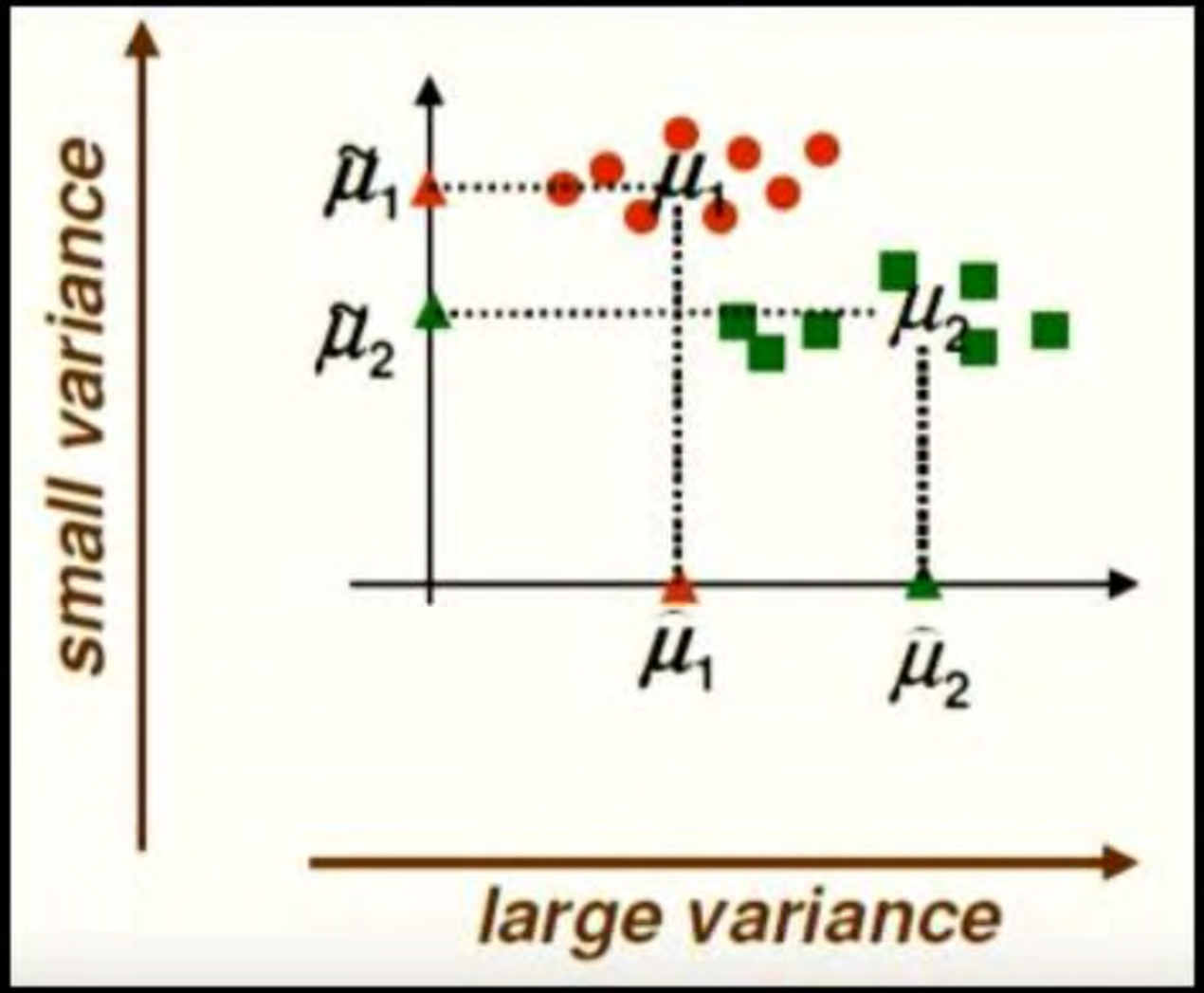
2) the variance of Projected points of different Classes shd be min.

Linear Discriminant Analysis

- Linear discriminant analysis (LDA) is based on Fisher's linear discriminant.

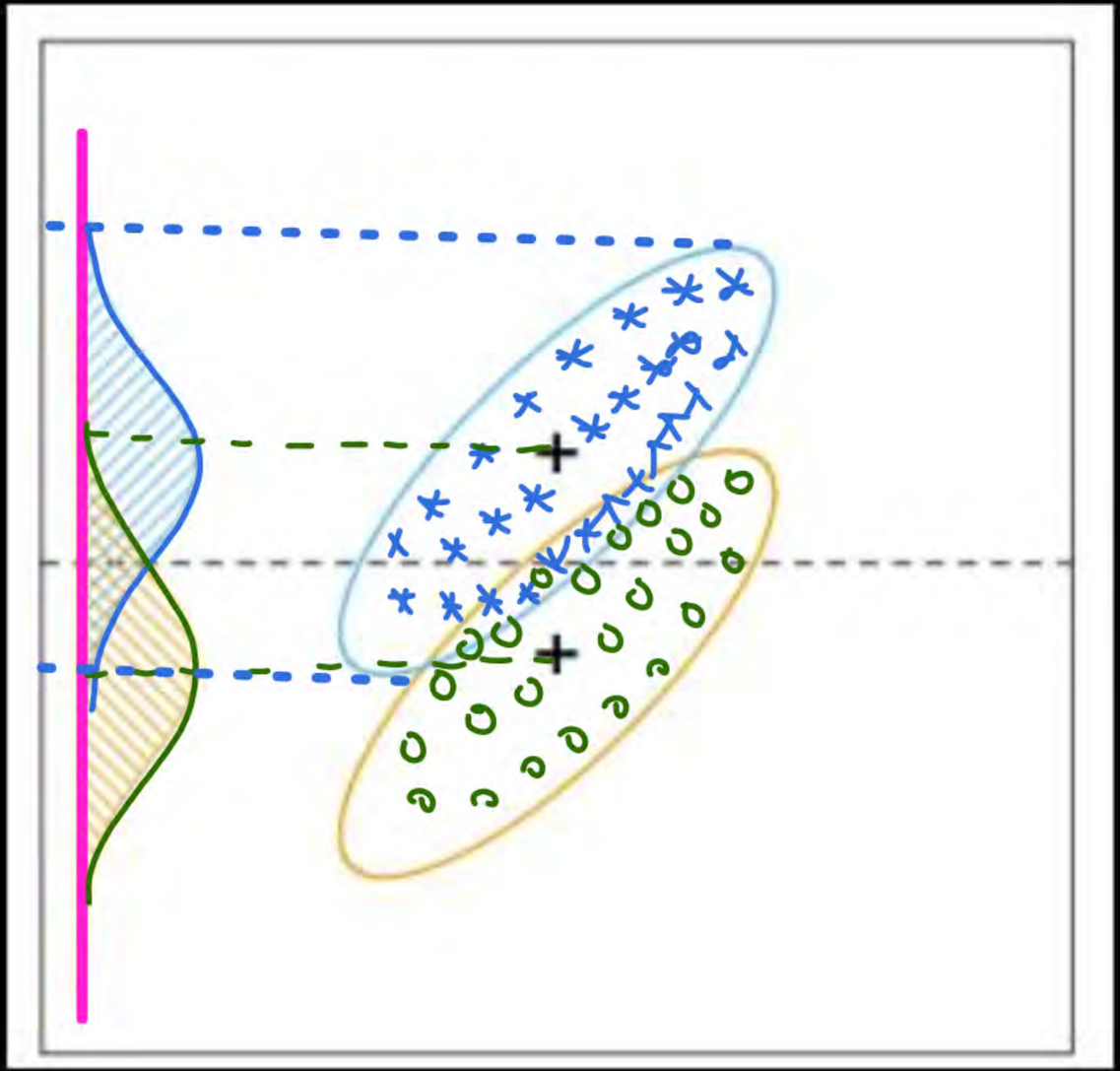


LDA – Fisher Linear Discriminant



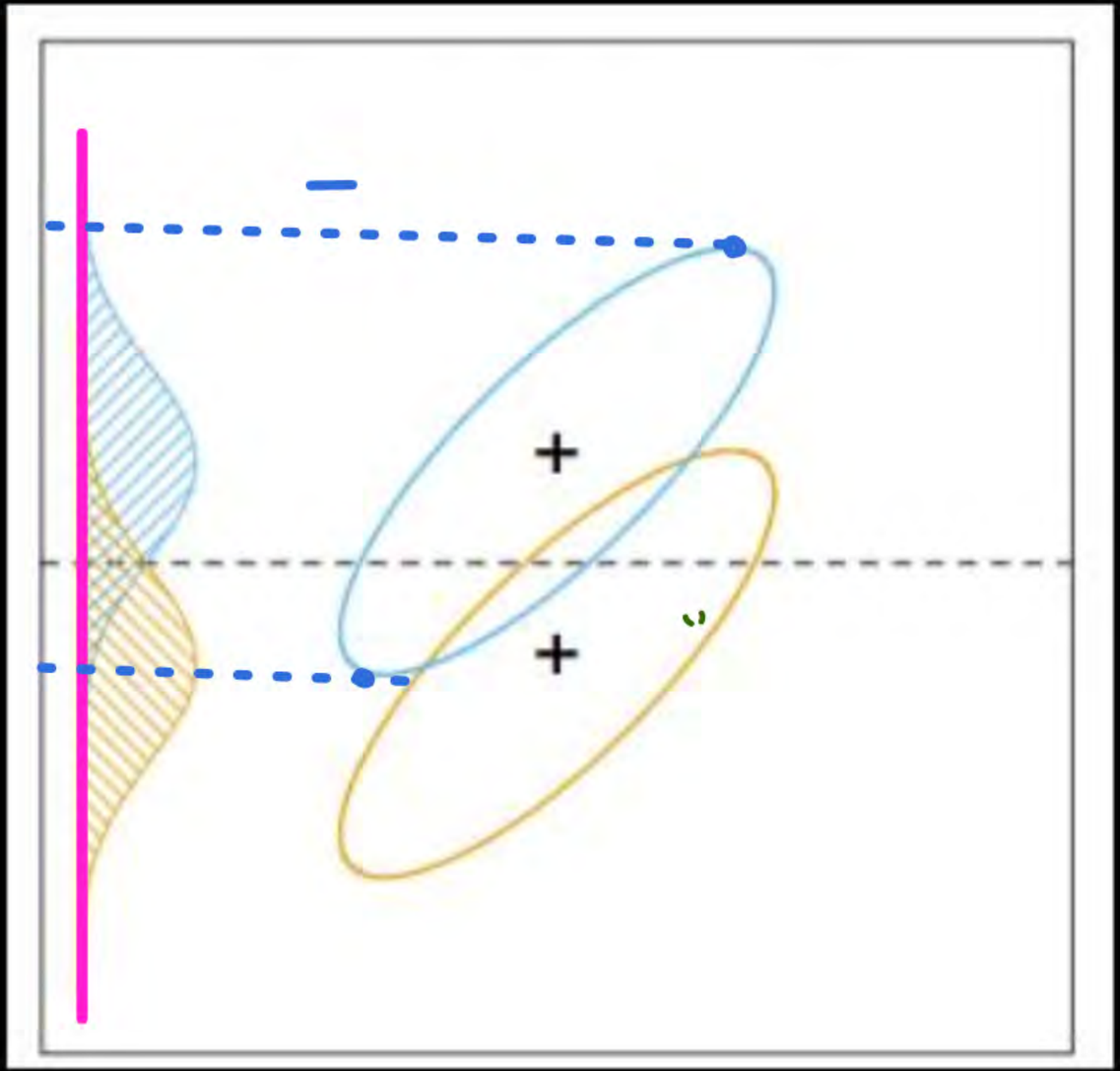
Linear Discriminant Analysis

What is a LDA ?



Linear Discriminant Analysis

What is a LDA ?



Linear Discriminant Analysis

What is a LDA ?

⇒ So for better projection we need ⇒

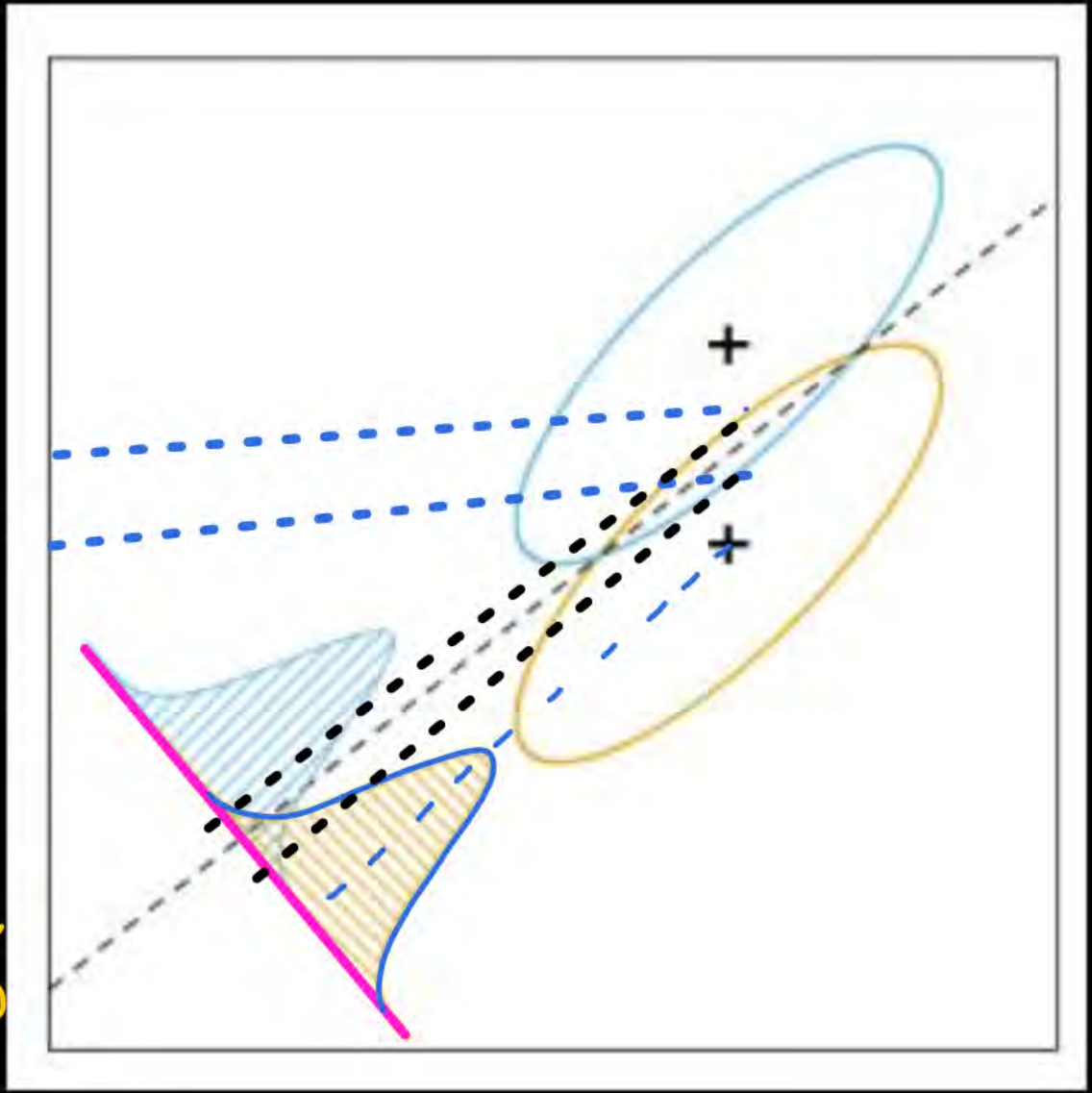
1) $\max \left[\tilde{\mu}_1 - \tilde{\mu}_0 \right]^2$

$\tilde{\mu}_1$ ⇒ Projected mean of class 1 Points

$\tilde{\mu}_0$ ⇒ Projected mean of class 0 Points

2) $\min \; \tilde{s}_1^2$ ⇒ Variance of Projected

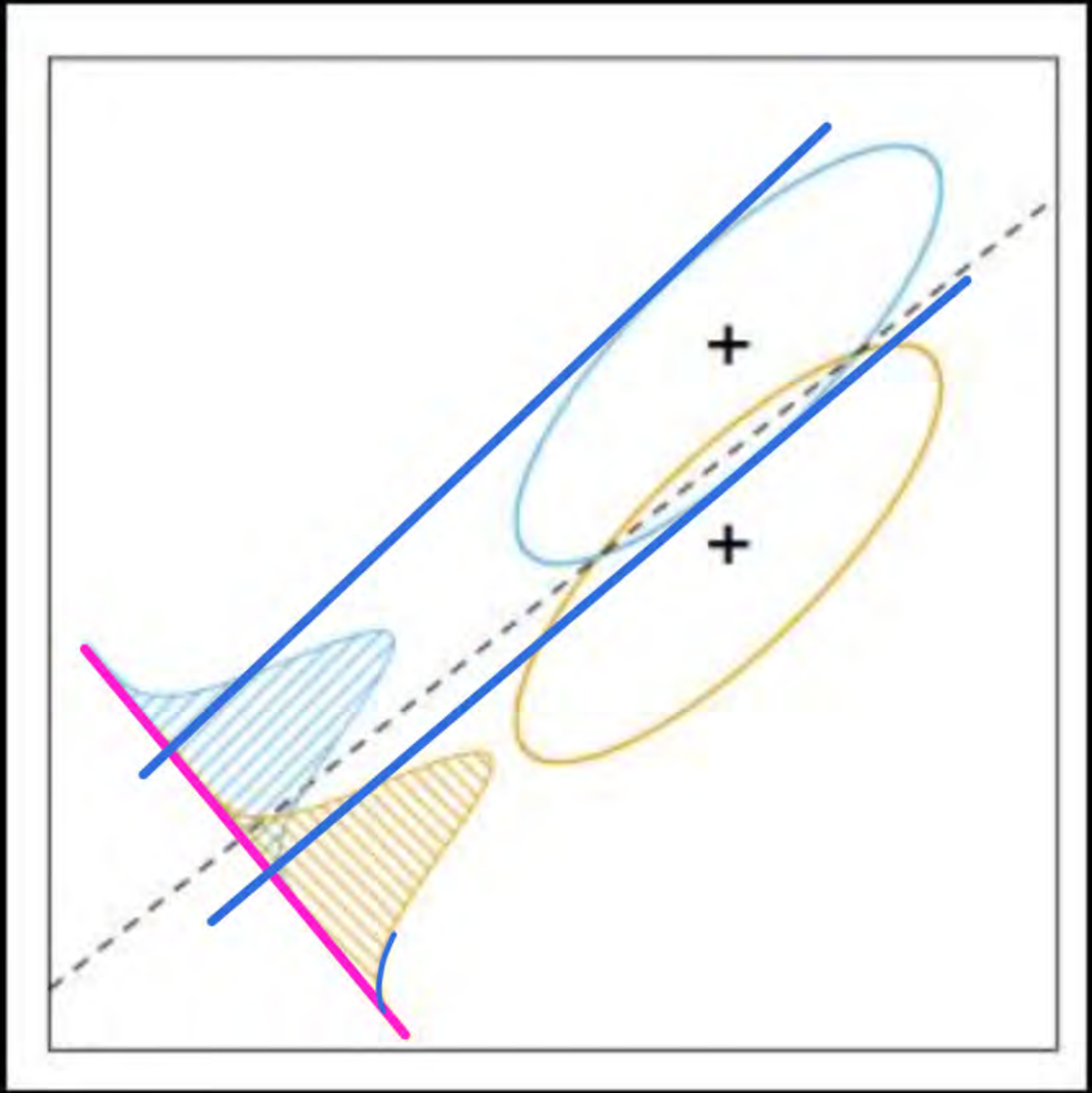
$\tilde{s}_0^2$ ⇒ Points of class 1 and 0



Linear Discriminant Analysis

What is a LDA ?

⇒ So for better projection we need ⇒



So we need to find ω vector

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_d \end{bmatrix}$$

So objective fxn $\Rightarrow$ $\max \left[\dfrac{(\tilde{\mu}_1 - \tilde{\mu}_0)^2}{S_0^2 + S_1^2} \right]$

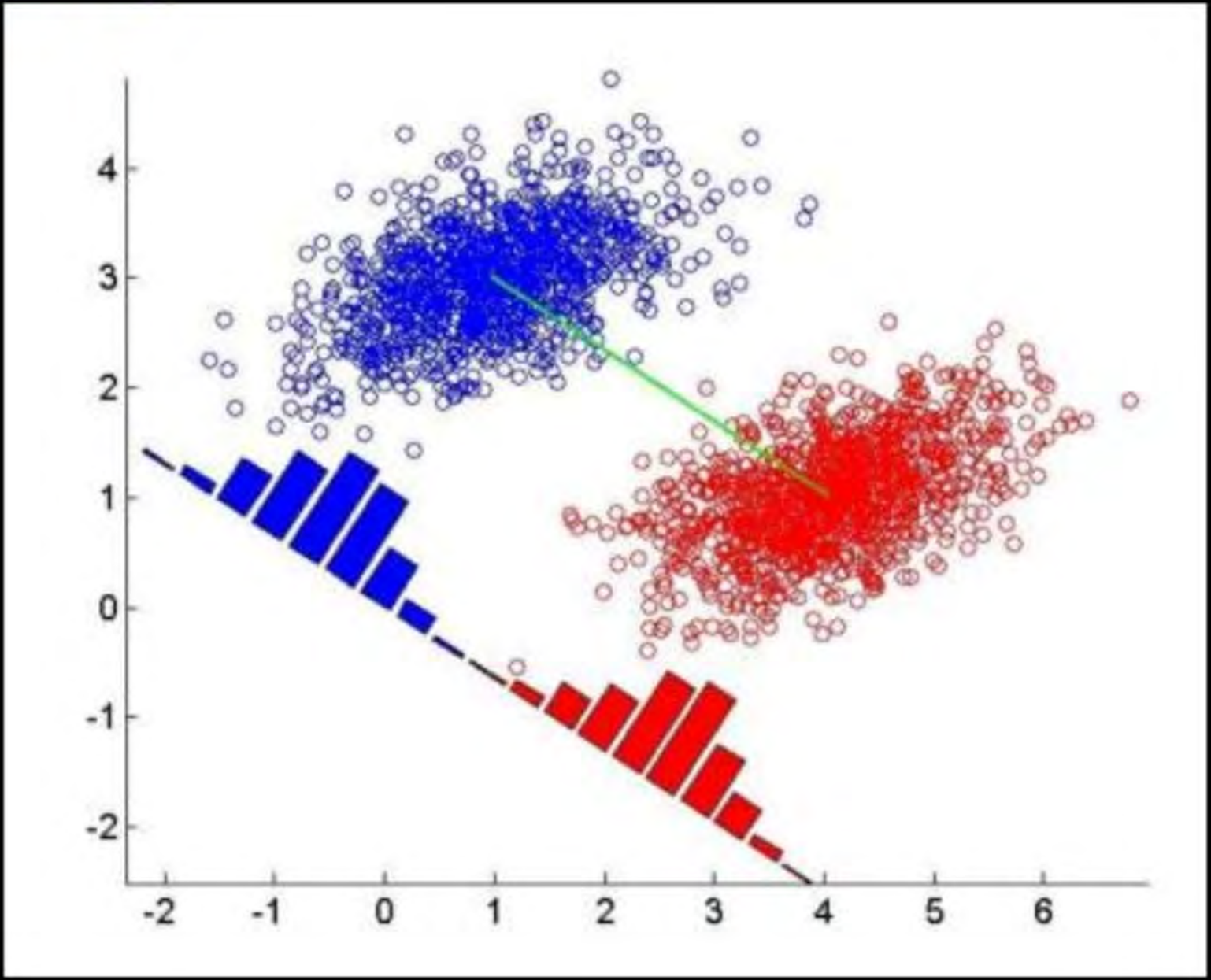
we will maximize gap b/w mean

we will min the S_0^2, S_1^2

What is a discriminant function

- Assumptions of LDA
- Normal Distribution: The features follow a Gaussian distribution.
- Equal Covariance: The covariance of every class is the same.
- Linearity: The relationships between the features are linear.

Linear Discriminant Analysis



LDA – Fisher Linear Discriminant

- To create this new axis and reduce dimensionality, LDA follows these criteria:
- Maximize the distance between the Projected means of two classes.
- Minimize the variance within individual classes.

Projected

LDA – Fisher Linear Discriminant

- How to find the projection of any point on a line …

data

X $\quad Y \Rightarrow$

→ Class 1 Points

$$\begin{matrix} x_1^1 & x_2^1 \\ x_1^2 & x_2^2 \\ \vdots & \vdots \\ x_1^D & x_2^D \end{matrix}$$

N_1 points

→ Class 0 Points

N_2 points

LDA – Fisher Linear Discriminant

Now, Class 1 points to be projected

Projected Points

$\omega^T x_i \Rightarrow (\omega^T)_{1\times d}(x_i)_{d\times 1} \Rightarrow (1\times 1)$

$\Rightarrow \tilde{\mu}_1 \Rightarrow$ mean of Projected Points

$\frac{1}{N_1}\sum_{i=1}^{N_1} \omega^T x_i$

$\Rightarrow \omega^T \underbrace{\sum_{i=1}^{N_1} \frac{x_i}{N_1}}$

$\Rightarrow \omega^T (\mu_1)$

LDA – Fisher Linear Discriminant

$$\mu_1 = \frac{1}{N_1}\sum_{i=1}^{N_1} x_i^{\circ}$$

mean value of class 1 points in original feature space

$$\begin{bmatrix} x_1^1 & x_2^1 & \cdots & x_{N_1}^1 \\ x_1^2 & x_2^2 & \cdots & x_{N_1}^2 \\ x_1^3 & x_2^3 & \cdots & \\ \vdots & \vdots & \cdots & \\ x_1^D & x_2^D & & x_{N_1}^D \end{bmatrix}$$

$$\Rightarrow \mu_1 = \begin{bmatrix} \frac{1}{N_1}\sum_{i=1}^{N_1} x_i^1 \\ \frac{1}{N_1}\sum_{i=1}^{N_1} x_i^2 \\ \vdots \\ \frac{1}{N_1}\sum_{i=1}^{N_1} x_i^D \end{bmatrix}$$

So $\tilde{\mu}_1 = \underbrace{w^T}_{1\times d} \underbrace{\mu_1}_{d\times 1}$

Projected mean of class 1

Original mean location of class 1

Similarly for class 0 points

$$\tilde{\mu}_0 = (\underbrace{w^T}_{1\times d} \underbrace{\mu_0}_{d\times 1})$$

$$\mu_0 = \frac{1}{N_2} \sum_{i=1}^{N_2} x_i^0$$

Now $\tilde{\mu}_1 - \tilde{\mu}_0 = w^T \mu_1 - w^T \mu_0$

$$= \overbrace{w^T \underbrace{(\mu_1 - \mu_0)}_{\text{gap b/w the mean location in original feature space}}} \Rightarrow \text{(Single value)}$$

$$(\tilde{\mu}_1 - \tilde{\mu}_0)^2 \Rightarrow \left(w^T(\mu_1 - \mu_0)\right)\left(w^T(\mu_1 - \mu_0)\right)^T$$

$$\Rightarrow \underbrace{w^T(\mu_1 - \mu_0)(\mu_1 - \mu_0)^T w}$$

$$(AB)^T = B^T A^T$$

$$(\tilde{u}_1 - \tilde{u}_0)^2 = \check{w}^T_{1\times d} (u_1 - u_0)_{d\times 1} \, (u_1 - u_0)^T_{1\times d} (\check{w})_{d\times 1} \Rightarrow \text{(Single value)}$$

Original data
Class mean values
available if
training data is
known

The $(u_1 - u_0)_{d\times 1} \, (u_1 - u_0)^T_{1\times d} \Rightarrow S_B(d\times d)$

Scatter matrix
between classes.

$$\text{So } (\tilde{\mu}_1 - \tilde{\mu}_D)^2 = (\omega^T S_B \omega)$$

$S_1^2 \Rightarrow$ Variance of Projected points of class 1

$$\frac{1}{N_1} \sum_{i=1}^{N_1} \left(\omega^T x_i^0 - \tilde{\mu}_1\right)^2 \Rightarrow \left(\omega^T x_i^0 - \tilde{\mu}_1\right)$$

Known

$$\left(\omega^T x_i^0 - \omega^T \mu_1\right)$$

Known

$$\Rightarrow \omega^T (x_i^0 - \mu_1)$$

So if we use all points of class 1

$$X_1 \Rightarrow \begin{bmatrix} x_1^1 & x_2^1 & , & x_{N_1}^1 \\ x_1^2 & x_2^2 & & x_{N_1}^2 \\ \vdots & \vdots & & \vdots \\ x_1^D & x_2^D & & x_{N_1}^D \end{bmatrix}$$

$$\mu_1 = \begin{bmatrix} \\ \\ \end{bmatrix}_{d \times 1}$$

$$\overset{1 \times d}{\Uparrow}$$

$$\omega^T (X_1 - \mu_1) \rightarrow d \times N_1$$

$$\Downarrow$$

$$\begin{bmatrix} \omega^T(x_1 - \mu_1) & \omega^T(x_2 - \mu_1) & \cdots & \omega^T(x_{N_1} - \mu_1) \end{bmatrix}_{1 \times N_1}$$

So $S_1^2 \Rightarrow \left(\omega^T \left[X_1 - \mu_1\right]\right)\left(\omega^T (X_1 - \mu_1)\right)^T$

$$\Rightarrow S_1^2 \Rightarrow \omega^T \underbrace{\left[X_1 - \mu_1\right]\left[X_1 - \mu_1\right]^T}_{\text{Known}} \omega$$

Similarly

$$S_0^2 \Rightarrow \omega^T \overbrace{\left[X_0 - \mu_0\right]\left[X_0 - \mu_0\right]^T}^{\text{Known}} \omega$$

So $s_1^2 + s_0^2 \Rightarrow \omega^T (x_1 - \mu_1)(x_1 - \mu_1)^T \omega + \omega^T (x_0 - \mu_0)(x_0 - \mu_0)^T \omega$

$$\Rightarrow \omega^T \underbrace{\left[(x_1 - \mu_1)(x_1 - \mu_1)^T + (x_0 - \mu_0)(x_0 - \mu_0)^T \right]}_{S_W} \omega$$

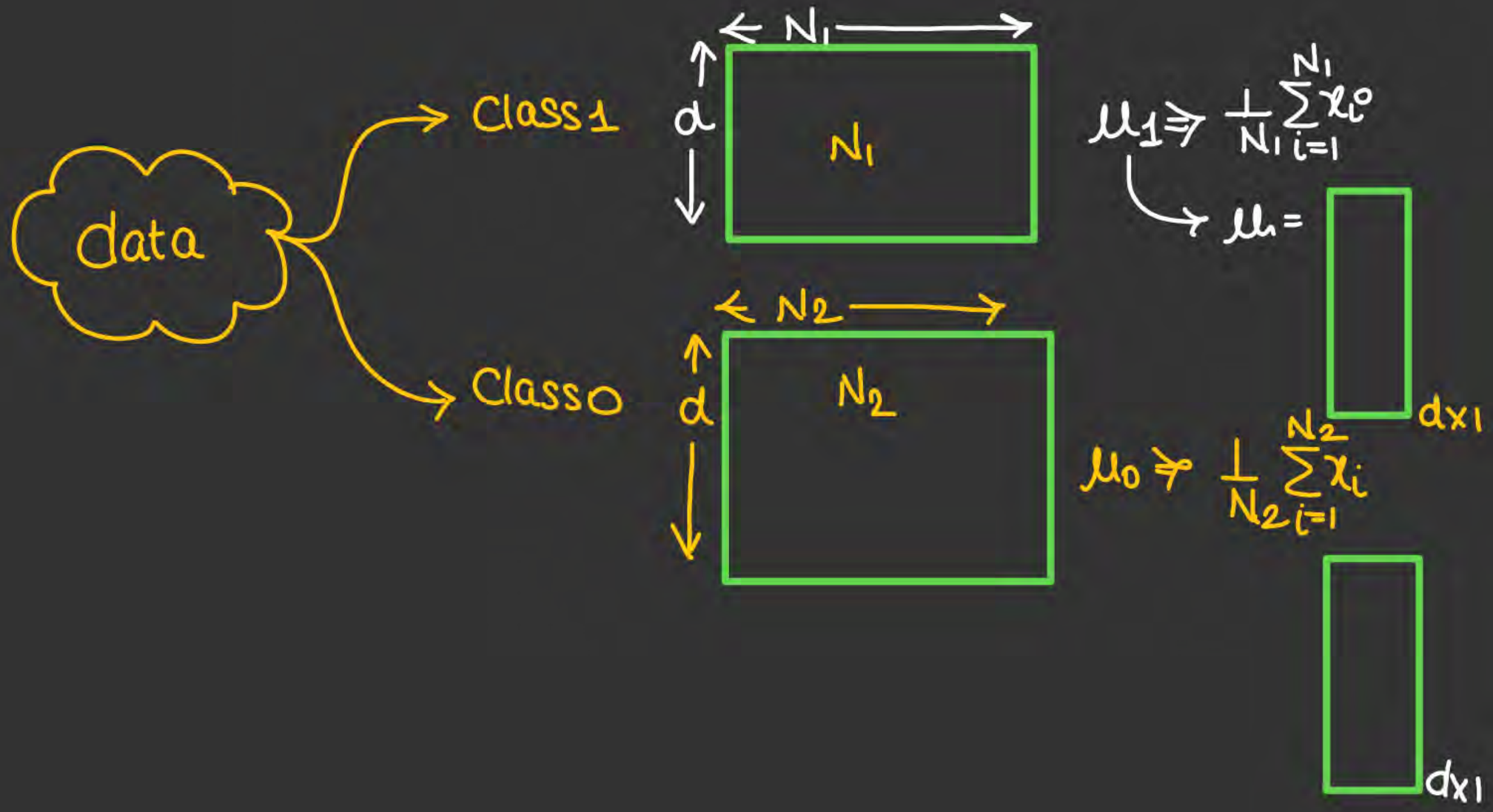
$$\Rightarrow (\omega^T S_W \omega)$$

$\rightarrow$ Scatter matrix within class.

So we have to find w to maximize $\Rightarrow \dfrac{w^T S_B w}{w^T S_W w}$

$$S_B \Rightarrow (\mu_1 - \mu_0)(\mu_1 - \mu_0)^T$$

$$S_W \Rightarrow (X_1 - \mu_1)(X_1 - \mu_1)^T + (X_0 - \mu_0)(X_0 - \mu_0)^T$$


data
Class 1
Class 0
N1
d
N1
N2
d
N2
$\mu_1 \Rightarrow \frac{1}{N_1}\sum_{i=1}^{N_1} x_i$
$\mu_1 =$
dx1
$\mu_0 \Rightarrow \frac{1}{N_2}\sum_{i=1}^{N_2} x_i$
dx1

Projected points mean

$$\tilde{\mu}_1 = (\omega^T \mu_1) \rightarrow \text{Single}$$

$$\tilde{\mu}_0 = (\omega^T \mu_0) \rightarrow \text{Value}$$

$$\Rightarrow (\tilde{\mu}_1 - \tilde{\mu}_0)^2$$

$$\Rightarrow \left(\omega^T \underbrace{(\mu_1 - \mu_0)(\mu_1 - \mu_0)^T}\omega\right)$$

$$\Rightarrow (\omega^T S_B \omega) \Rightarrow \text{Single Value}$$

$$S_1^2 \Rightarrow \sum_{i=1}^{N_1} \left(\omega^T x_i - \tilde{u}_1\right)^2$$

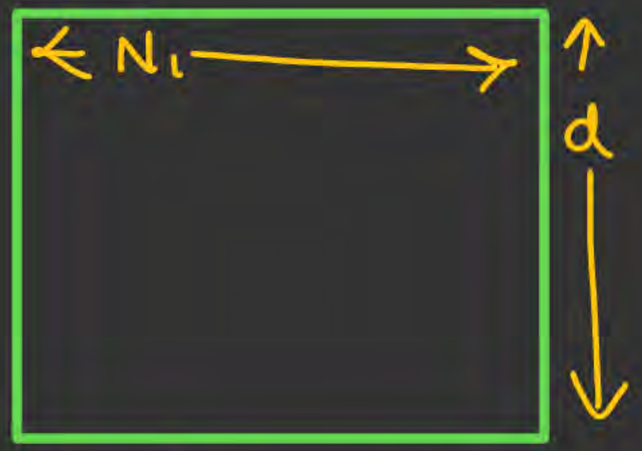
$$\Rightarrow \omega^T (x_i^0 - u_1)$$

$$\left(\omega^T (X_1 - \mu_1)(X_1 - \mu_1)^T \omega\right)$$

$\omega^T \to 1 \times d$, $(X_1 - \mu_1) \to d \times N_1$, $(X_1 - \mu_1)^T \to N_1 \times d$, $\omega \to d \times 1$

$$S_0^2 \Rightarrow \left(\omega^T (X_0 - \mu_0)(X_0 - \mu_0)^T \omega\right)$$

Class 1 points

$X_1 =$ 

$$\omega^T (X_1 - \mu_1) = \left[\omega^T (x_1 - \mu_1) \;\; \omega^T (x_2 - \mu_1) \; \cdots\right]$$

$$S_W = (x_1 - \mu_1)(x_1 - \mu_1)^T + (x_0 - \mu_0)(x_0 - \mu_0)^T$$

$$\max \left(\frac{w^T S_B w}{w^T S_W w} \right) \Rightarrow$$

Fisher discriminant fxn.

(NAT)

Steps

- Compute the d-dimensional mean vectors for the different classes from the dataset.
- Compute the scatter matrices (in-between-class and within-class scatter matrix).
- Compute the eigenvectors (e_1, e_2, ..., e_d) and corresponding eigenvalues (λ_1, λ_2, ..., λ_d) for the scatter matrices.
- Sort the eigenvectors by decreasing eigenvalues and choose k eigenvectors with the largest eigenvalues to form a d × k dimensional matrix W (where every column represents an eigenvector).
- Use this d × k eigenvector matrix to transform the samples onto the new subspace. This can be summarized by the matrix multiplication: Y = X × W (where X is a n × d-dimensional matrix representing the n samples, and y are the transformed n × k-dimensional samples in the new subspace).

LDA – Fisher Linear Discriminant

. LDA Algorithm

1. **Compute the mean vectors** of each class.
2. **Compute the scatter matrices** S_W and S_B.
3. **Compute the eigenvectors and eigenvalues** for the matrix $S_W^{-1} S_B$.
4. **Sort the eigenvectors** by decreasing eigenvalues and choose the top k eigenvectors to form a matrix W.
5. **Transform the samples** to the new space using W.

What is a discriminant function

- Linear Discriminant Analysis (LDA) is a statistical technique for categorizing data into groups. It identifies patterns in features to distinguish between different classes. For instance, it may analyze characteristics like size and color to classify fruits as apples or oranges. LDA aims to find a straight line or plane that best separates these groups while minimizing overlap within each class. By maximizing the separation between classes, it enables accurate classification of new data points.
- Linear discriminant analysis (LDA) is an approach used in supervised machine learning to solve multi-class classification problems. LDA separates multiple classes with multiple features through data dimensionality reduction.

Practise

1. What is the primary goal of Linear Discriminant Analysis (LDA)?

a) To maximize the variance of the data.

b) To minimize the covariance between classes.

c) To maximize the separability between different classes.

d) To find non-linear boundaries between classes.

Practise

2. Which assumption is NOT made by LDA?

a) The features follow a Gaussian distribution.

b) The covariance matrices of the classes are equal.

c) The relationships between the features are linear.

d) The features are independent of each other.

Practise

3. In LDA, the Within-Class Scatter Matrix (S_W) measures:

a) The variance within each class.

b) The variance between the classes.

c) The covariance of the entire dataset.

d) The overall mean vector of the data.

Practise

4. How is the Between-Class Scatter Matrix (S_B) computed in LDA?

a) By summing the covariance matrices of all classes.

b) By summing the outer product of the difference between the class means and the overall mean.

c) By calculating the mean of all data points.

d) By multiplying the within-class scatter matrix by its transpose.

Practise

5. Which of the following is a key difference between LDA and PCA?

a) LDA maximizes variance while PCA maximizes class separability.

b) PCA maximizes variance while LDA maximizes class separability.

c) Both LDA and PCA maximize class separability.

d) Both LDA and PCA maximize variance.

Practise

6. In the context of LDA, what is the primary purpose of eigenvectors and eigenvalues?

a) To find the principal components of the data.

b) To transform the data into a new space with reduced dimensions.

c) To maximize the covariance within each class.

d) To minimize the between-class scatter.

Practise

7. Which step comes first in the LDA algorithm?

a) Compute the eigenvectors and eigenvalues.

b) Transform the samples to the new space.

c) Compute the mean vectors of each class.

d) Sort the eigenvectors by decreasing eigenvalues.

Practise

8. LDA is particularly effective when:

a) The classes are linearly separable.

b) The classes are not linearly separable.

c) There are more features than samples.

d) The data follows a non-Gaussian distribution.

How to classify using the LDA

- **Classify the Samples**
- After transformation, the samples can be classified using a classification algorithm such as:
  - **Nearest Centroid Classifier**: Classify a new sample to the class whose mean vector is closest to the sample in the transformed space.
  - **Linear Discriminant Function**: Use the discriminant functions to assign samples to the class with the highest discriminant score.
  - **Other Classifiers**: Use other classifiers like k-Nearest Neighbors (k-NN) or support vector machines (SVM) on the reduced dimensionality data.

Practise

5. Consider the case where two classes follow Gaussian distribution which are centered at (−1, 2) and (1, 4) and have identity covariance matrix. Which of the following is the separating decision boundary?

(a) $y - x = 3$

(b) $x + y = 3$

(c) $x + y = 6$

(d) (b) and (c) are possible

e) None of these

(f) Can not be found from the given information

8. We discussed the use of MLE for the estimation of parameters of logistic regression model. We used which of the following assumptions to derive the likelihood function ?

a. independence among the class labels

b. independence among each training sample

c. independence among the parameters of the model

d. None of these

THANK - YOU